

PART-A

Name of the Student: GARVIL SETHI

Signature of the Student: *Garvil*

Roll No.: 230123082

Signature of the Invigilator: *AKH*

INSTRUCTIONS

1. Check that you have **10 printed pages**. The PART-A of the examination has **3 questions**, for a total of **20 points**.
2. Attempt **all questions**.
3. There is **no credit** for a solution if the appropriate work is not shown, even if the answer is correct.
4. Write your answers in this booklet and only in the **space marked for answers**. Any answer written in any space other than the designated space will **not be evaluated**.
5. The pages numbered **8 to 10** are kept as **additional pages**. If you cannot able to fit a complete answer in the space provided just after the question, you can use these three pages by referring the page number of additional page in the original space.
6. At the beginning of the examination, a supplementary sheet has been provided only for rough work. Should you need more, please ask the invigilator.
7. No question requires any clarification from the instructors or invigilators, even if a question has an error or incomplete data. The students are advised to write answer according to their understanding or write reasons for why it is not possible to solve partially or completely that question by citing errors/insufficient data.

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Question:	1	2	3	Total
Points:	5	6	9	20
Score:	5	2	9	16

QUESTIONS

1. (5 points) Let Y and ϵ are the response variable and the error in a regression problem, respectively. θ_1 and θ_2 are the two parameters. Let $y_i, i = 1, 2, 3$, be the observed values of the response variable. Consider,

$$Y_1 = \theta_1 + \theta_2 + \epsilon_1,$$

$$Y_2 = \theta_1 - 2\theta_2 + \epsilon_2,$$

$$Y_3 = 2\theta_1 - \theta_2 + \epsilon_3,$$

where $E(\epsilon_i) = 0$, and $V(\epsilon_i) = \sigma^2, i = 1, 2, 3; \sigma > 0$. Find the least square estimates (LSE) of θ_1 and θ_2 .

~~$$Y = \theta_1 + \theta_2 + \epsilon \quad \text{let } Y_4 = \frac{Y_1 + Y_3}{2} = \theta_1 - \frac{\theta_2}{2} + \epsilon_4, \quad \epsilon_4 = \frac{\epsilon_1 + \epsilon_3}{2}$$~~

~~$$Y = \theta_1 + \theta_2 + \epsilon \quad \text{let } Y_4 = \theta_1 - \frac{\theta_2}{2} + \epsilon_4, \quad \epsilon_4 = \frac{\epsilon_1 + \epsilon_3}{2}$$~~

$$Y = X\theta + \epsilon$$

$\begin{matrix} 3 \times 1 & & 3 \times 2 & 2 \times 1 & & 3 \times 1 \end{matrix}$

$$Y = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}, \quad X = \begin{bmatrix} 1 & 1 \\ 1 & -2 \\ 2 & -1 \end{bmatrix}, \quad \theta = \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix}, \quad \epsilon = \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \epsilon_3 \end{bmatrix}$$

$$\hat{\theta} = (X^T X)^{-1} X^T Y$$

$$X^T X = \begin{bmatrix} 1 & 1 & 2 \\ 1 & -2 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -2 \\ 2 & -1 \end{bmatrix} = \begin{bmatrix} 6 & -3 \\ -3 & 6 \end{bmatrix}$$

$$(X^T X)^{-1} = \frac{1}{36 - 9} \begin{bmatrix} 6 & 3 \\ 3 & 6 \end{bmatrix} = \frac{1}{27} \begin{bmatrix} 6 & 3 \\ 3 & 6 \end{bmatrix}$$

$$(X^T X)^{-1} X^T = \frac{1}{27} \begin{bmatrix} 6 & 3 \\ 3 & 6 \end{bmatrix} \begin{bmatrix} 1 & 1 & 2 \\ 1 & -2 & -1 \end{bmatrix}$$

$$= \frac{1}{27} \begin{bmatrix} 6 & 3 \\ 3 & 6 \end{bmatrix} \frac{1}{27} \begin{bmatrix} 9 & 0 & 9 \\ 9 & -9 & 0 \end{bmatrix}$$

$$(X^T X)^{-1} X^T y = \frac{1}{27} \begin{bmatrix} ay_1 + ay_3 \\ ay_1 - ay_2 \end{bmatrix} = \begin{bmatrix} \hat{\theta}_1 \\ \hat{\theta}_2 \end{bmatrix}$$

$$\therefore \hat{\theta}_1 = \frac{y_1}{3} + \frac{y_3}{3}$$

are the LSEs for θ_1 and θ_2

$$\hat{\theta}_2 = \frac{y_1}{3} - \frac{y_2}{3}$$

OS

2. Let U_1 and U_2 be two independent uniform random variables on the interval $(0, 1)$. Suppose that $X = (X_1, X_2, X_3)'$, where $X_1 = U_1$, $X_2 = U_2$, and $X_3 = U_1 + U_2$.

(a) (3 points) Compute the correlation matrix ρ of X .

(b) (3 points) Find the variance of first principal component based on the correlation matrix that you find in part (a).

$$(a) \rho = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} \quad \begin{aligned} r_{11} &= \text{var}(U_1) = \text{var}(X_1) \\ &= \text{var}(U_1) = \frac{1}{12} \\ r_{22} &= \text{var}(X_2) = \text{var}(U_2) = \frac{1}{12} \end{aligned}$$

$$\begin{aligned} r_{33} &= \text{var}(X_3) = \text{var}(U_1 + U_2) = \text{var}(U_1) + \text{var}(U_2) + 2\text{cov}(U_1, U_2) \\ &= \frac{1}{12} + \frac{1}{12} + 0 = \frac{1}{6} \end{aligned} \quad \begin{array}{l} \text{Since } U_1, U_2 \\ \text{are independent} \end{array}$$

$$r_{12} = \frac{\text{cov}(X_1, X_2)}{\sqrt{\text{var}(X_1)} \sqrt{\text{var}(X_2)}} = \frac{\text{cov}(U_1, U_2)}{\sqrt{\text{var}(U_1)} \sqrt{\text{var}(U_2)}} = 0 = r_{21}$$

$$\begin{aligned} r_{13} &= \frac{\text{cov}(U_1, U_1 + U_2)}{\sqrt{\text{var}(U_1)} \sqrt{\text{var}(U_1 + U_2)}} = \frac{\text{cov}(U_1, U_1) + \text{cov}(U_1, U_2)}{\sqrt{\frac{1}{12}} \times \sqrt{\frac{1}{6}}} \\ &= \frac{\text{var}(U_1) + 0}{\frac{1}{6\sqrt{2}}} = 6\sqrt{2} \times \frac{1}{12} = \frac{1}{\sqrt{2}} = r_{31} \end{aligned}$$

$$\therefore U_1 \sim U_2 \therefore r_{23} = r_{32} = r_{13} = \frac{1}{\sqrt{2}}$$

$$\therefore \rho = \begin{bmatrix} 1/12 & 0 & 1/\sqrt{2} \\ 0 & 1/12 & 1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} & 1/6 \end{bmatrix}$$

01

$$(b) \det(J - \lambda I) = 0 \Rightarrow \det \begin{bmatrix} 1/12 - \lambda & 0 & 1/\sqrt{2} \\ 0 & 1/12 - \lambda & 1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} & 1/6 - \lambda \end{bmatrix} = 0$$

$$\Rightarrow \left(\frac{1}{12} - \lambda\right) \left[\left(\frac{1}{12} - \lambda\right) \left(\frac{1}{6} - \lambda\right) - \frac{1}{2} \right]$$

$$+ \frac{1}{\sqrt{2}} \left(-\frac{1}{\sqrt{2}} \left(\frac{1}{12} - \lambda\right) \right) = 0$$

$$\Rightarrow \left(\frac{1}{12} - \lambda\right) \left(\frac{1}{72} + \lambda^2 - \frac{\lambda}{4}\right) - \frac{1}{24} + \frac{\lambda}{2} = 0$$

$$\Rightarrow \frac{-35}{864} + \frac{35}{72} \lambda + \frac{\lambda^2}{12} - \lambda^3 - \frac{\lambda}{48} + \frac{\lambda^2}{4} - \frac{1}{24} + \frac{\lambda}{2} = 0$$

$$\Rightarrow \lambda^3 - \frac{\lambda^2}{3} - \frac{139}{144} \lambda + \frac{71}{864} = 0 \rightarrow \textcircled{1}$$

Variance of 1st principal component $y_1 = \text{var}(y_1) = \lambda_1$
which is the largest eigenvalue of J .

which is the largest root of eqn ①

can be calculated using spectral radius $\lambda_1 = \sqrt{\text{eigenvalue of } (J^T J)^{-1}}$

01

3. Let the matrix of distance be

	1	2	3	4
1	0			
2	1	0		
3	11	2	0	
4	5	3	3.5	0

(a) (6 points) Cluster the four items using each of the following procedures:

- Complete linkage hierarchical procedure.
- Average linkage hierarchical procedure.

(b) (3 points) Draw the dendrograms and compare the results obtained using previous two procedures.

(a) i. complete linkage

minimum value in distance matrix = $d_{12} = d_{21} = 1$

so clusters: $(1,2)$, $(3,4)$

new distance matrix:

$$d_{(1,2),3} = \max(d_{13}, d_{23})$$

$$= \max(11, 2)$$

$$= 11$$

	(1,2)	3	4
(1,2)	0		
3	11	0	
4	5	3.5	0

$$d_{(1,2),4} = \max(d_{41}, d_{42})$$

$$= \max(5, 3)$$

$$= 5$$

minimum value = 3.5

so clusters: $(1,2)$, $(3,4)$

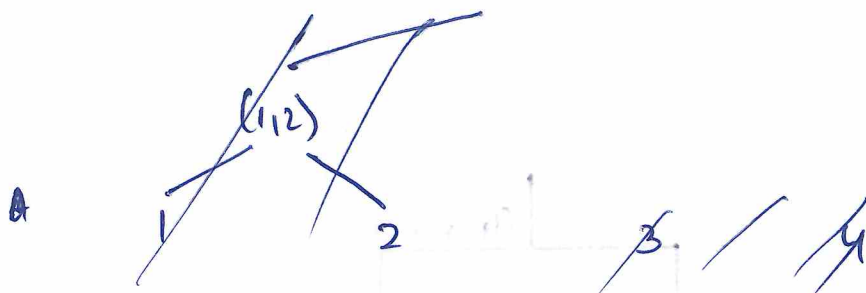
new distance matrix:

$$d_{(1,2),(3,4)} = \max(d_{31}, d_{32}, d_{41}, d_{42})$$

$$= \max(11, 2, 5, 3)$$

$$= 11$$

	(1,2)	(3,4)
(1,2)	0	
(3,4)	11	0



clustering:

$$(1), (2), (3), (4) \rightarrow (1,2), (3), (4) \rightarrow (1,2), (3,4) \rightarrow (1,2,3,4)$$

ii. average linkage

minimum value in distance matrix = $d_{21} = 1$

so clusters: $(1,2), (3), (4)$

new distance matrix:

$$d_{(1,2),3} = \frac{d_{(1,3)} + d_{(2,3)}}{2 \times 1} = \frac{11 + 2}{2} = 6.5$$

$$d_{(1,2),4} = \frac{d_{(1,4)} + d_{(2,4)}}{2 \times 1} = \frac{5 + 3}{2} = 4$$

$$\begin{matrix} & \begin{matrix} (1,2) & 3 & 4 \end{matrix} \\ \begin{matrix} (1,2) \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & & \\ 6.5 & 0 & \\ 4 & \textcircled{3.5} & 0 \end{bmatrix} \end{matrix}$$

minimum value in distance matrix = 3.5

so clusters: $(1,2), (3,4)$

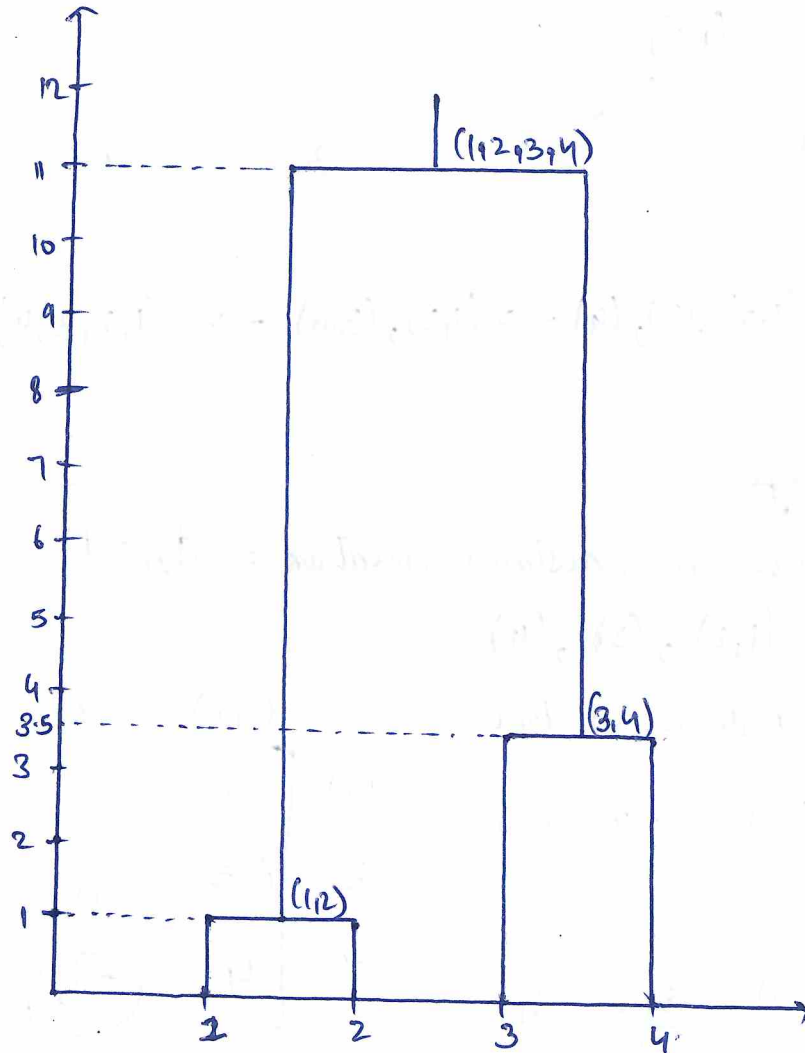
new distance matrix:

$$\begin{aligned} d_{(1,2),(3,4)} &= \frac{d_{(1,3)} + d_{(1,4)} + d_{(2,3)} + d_{(2,4)}}{2 \times 2} \\ &= \frac{11 + 2 + 5 + 3}{4} = \frac{21}{4} = 5.25 \end{aligned}$$

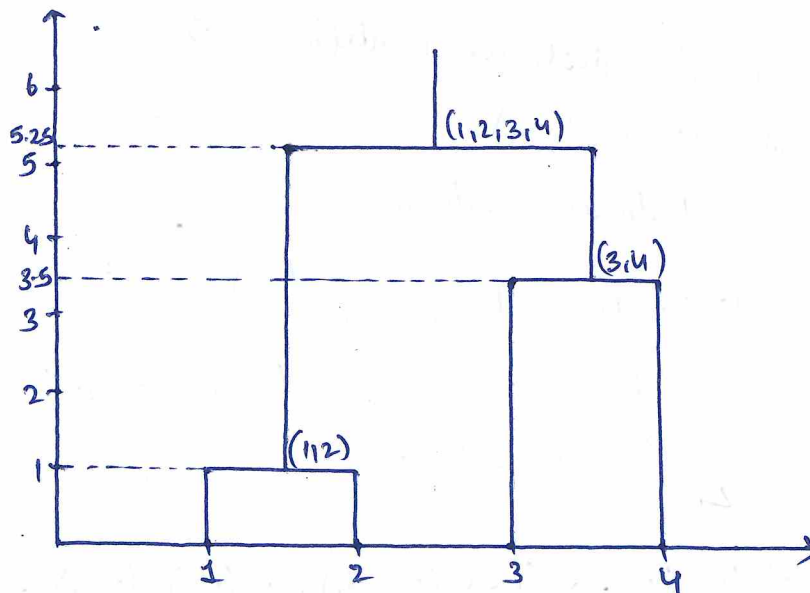
$$\begin{matrix} & \begin{matrix} (1,2) & (3,4) \end{matrix} \\ \begin{matrix} (1,2) \\ (3,4) \end{matrix} & \begin{bmatrix} 0 & \\ 5.25 & 0 \end{bmatrix} \end{matrix}$$

$\therefore$ clustering: $(1), (2), (3), (4) \rightarrow (1,2), (3), (4) \rightarrow (1,2), (3,4) \rightarrow (1,2,3,4)$

(b) Dendrogram for complete linkage:



Dendrogram
for average
linkage



OBSERVATIONS : Using both the linkage methods, same clusters have been formed in order but the final cluster gets formed at 11 for complete linkage and at 5.25 for average linkage.

03

1. The following are the main features of the Indian Constitution:
(a) It is a single document.
(b) It is a written constitution.
(c) It is a rigid constitution.
(d) It is a federal constitution.
(e) It is a democratic constitution.
(f) It is a secular constitution.
(g) It is a constitutional monarchy.
(h) It is a parliamentary system.
(i) It is a unitary system.
(j) It is a free state.